

MATRIX OPERATIONS USING FUNCTIONS AND 2D ARRAYS

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Aim :

To implement Matrix Addition, Matrix Multiplication, and Matrix Transpose using functions and 2D arrays in C.

Algorithm :

Matrix Addition

- Read rows and columns.
- Input elements of two matrices.
- Add corresponding elements.
- Display result.

Matrix Multiplication

Check if multiplication is possible.

- Multiply rows and columns.
- Store result matrix.
- Display output.

Transpose

- Swap rows with columns.
- Store transposed matrix.
- Display result.

Program

```
#include <stdio.h>
```

```
#define MAX 10
```

```
// Function to input matrix
```

```
void inputMatrix(int matrix[MAX][MAX], int rows, int cols)
```

```
{
```

```
    int i, j;
```

```
    for(i = 0; i < rows; i++)
```

```
    {
```

```
        for(j = 0; j < cols; j++)
```

```
        {
```

```
            printf("Enter element [%d][%d]: ", i, j);
```

```
            scanf("%d", &matrix[i][j]);
```

```
        }
```

```
    }
```

```
}
```

```
// Function to display matrix
```

```
void displayMatrix(int matrix[MAX][MAX], int rows, int cols)
{
    int i, j;

    for(i = 0; i < rows; i++)
    {
        for(j = 0; j < cols; j++)
        {
            printf("%d\t", matrix[i][j]);
        }
        printf("\n");
    }
}
```

// Function for matrix addition

```
void addMatrix(int A[MAX][MAX], int B[MAX][MAX],
               int result[MAX][MAX], int rows, int cols)
{
    int i, j;

    for(i = 0; i < rows; i++)
    {
        for(j = 0; j < cols; j++)
        {
            result[i][j] = A[i][j] + B[i][j];
        }
    }
}
```



```
for(i = 0; i < rows; i++)  
{  
    for(j = 0; j < cols; j++)  
    {  
        transpose[j][i] = matrix[i][j];  
    }  
}  
}
```

```
int main()  
{  
    int A[MAX][MAX], B[MAX][MAX];  
    int sum[MAX][MAX], product[MAX][MAX];  
    int transpose[MAX][MAX];  
  
    int r1, c1, r2, c2;  
  
    printf("Enter rows and columns of Matrix A: ");  
    scanf("%d%d", &r1, &c1);  
  
    printf("Enter rows and columns of Matrix B: ");  
    scanf("%d%d", &r2, &c2);  
  
    printf("\nEnter elements of Matrix A:\n");  
    inputMatrix(A, r1, c1);  
  
    printf("\nEnter elements of Matrix B:\n");
```

```
inputMatrix(B, r2, c2);

// Matrix Addition
if(r1 == r2 && c1 == c2)
{
    addMatrix(A, B, sum, r1, c1);

    printf("\nMatrix Addition Result:\n");
    displayMatrix(sum, r1, c1);
}
else
{
    printf("\nMatrix Addition not possible!\n");
}

// Matrix Multiplication
if(c1 == r2)
{
    multiplyMatrix(A, B, product, r1, c1, c2);

    printf("\nMatrix Multiplication Result:\n");
    displayMatrix(product, r1, c2);
}
else
{
    printf("\nMatrix Multiplication not possible!\n");
}
```

```

// Transpose of Matrix A

transposeMatrix(A, transpose, r1, c1);

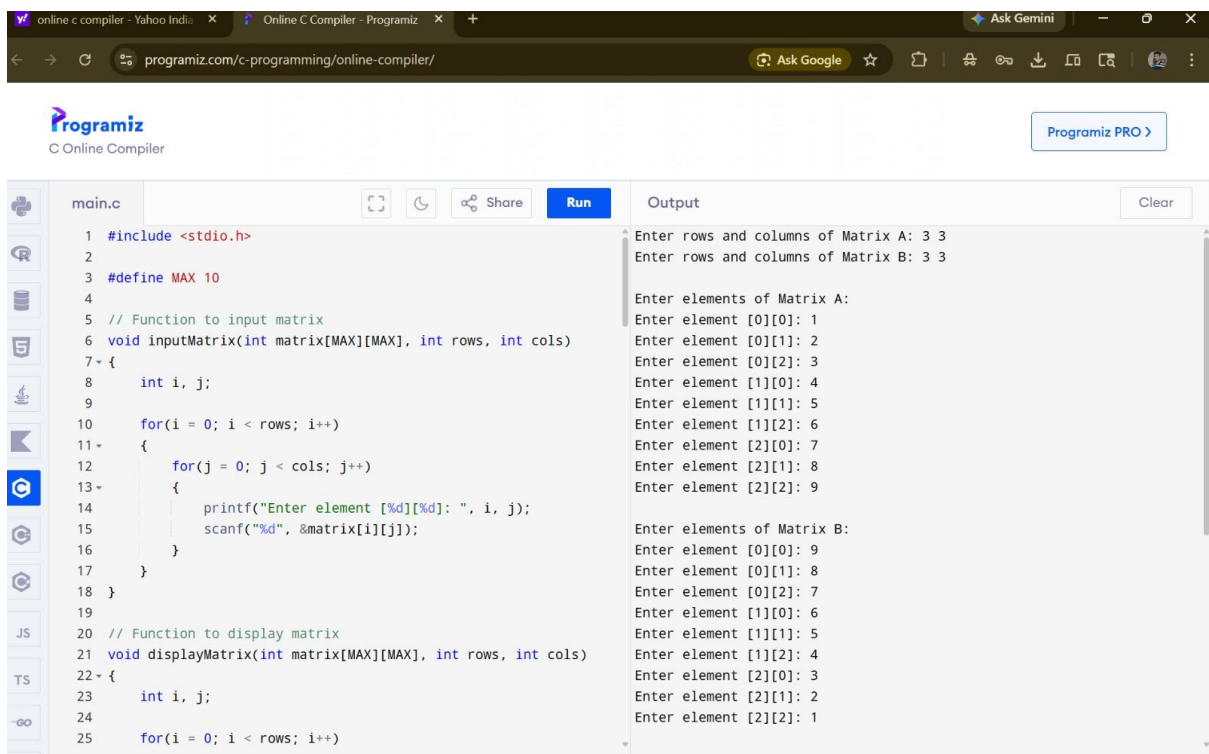
printf("\nTranspose of Matrix A:\n");

displayMatrix(transpose, c1, r1);

return 0;
}

```

Output



The screenshot shows the Programiz Online C Compiler interface. The code in the editor is as follows:

```

1 #include <stdio.h>
2
3 #define MAX 10
4
5 // Function to input matrix
6 void inputMatrix(int matrix[MAX][MAX], int rows, int cols)
7 {
8     int i, j;
9
10    for(i = 0; i < rows; i++)
11    {
12        for(j = 0; j < cols; j++)
13        {
14            printf("Enter element [%d][%d]: ", i, j);
15            scanf("%d", &matrix[i][j]);
16        }
17    }
18 }
19
20 // Function to display matrix
21 void displayMatrix(int matrix[MAX][MAX], int rows, int cols)
22 {
23     int i, j;
24
25     for(i = 0; i < rows; i++)

```

The output window shows the following text:

```

Enter rows and columns of Matrix A: 3 3
Enter rows and columns of Matrix B: 3 3

Enter elements of Matrix A:
Enter element [0][0]: 1
Enter element [0][1]: 2
Enter element [0][2]: 3
Enter element [1][0]: 4
Enter element [1][1]: 5
Enter element [1][2]: 6
Enter element [2][0]: 7
Enter element [2][1]: 8
Enter element [2][2]: 9

Enter elements of Matrix B:
Enter element [0][0]: 9
Enter element [0][1]: 8
Enter element [0][2]: 7
Enter element [1][0]: 6
Enter element [1][1]: 5
Enter element [1][2]: 4
Enter element [2][0]: 3
Enter element [2][1]: 2
Enter element [2][2]: 1

```

Run

Output

Clear

Enter element [0][2]: 7
Enter element [1][0]: 6
Enter element [1][1]: 5
Enter element [1][2]: 4
Enter element [2][0]: 3
Enter element [2][1]: 2
Enter element [2][2]: 1

Matrix Addition Result:
10 10 10
10 10 10
10 10 10

Matrix Multiplication Result:
30 24 18
84 69 54
138 114 90

Transpose of Matrix A:
1 4 7
2 5 8
3 6 9

=== Code Execution Successful ===

Result:

Thus, the program for Matrix Addition, Matrix Multiplication, and Matrix Transpose was implemented successfully using functions and 2D arrays in C. The use of functions improved modularity and code reusability, while 2D arrays helped in efficient matrix operations and data handling.